Porthole user guide

Porthole is **a variable fisheye and defish lens warp for Resolume Arena and Avenue**, as an FFGL effect.

Rather than drawing a bulge, it **re-photographs the picture through a different lens**: every output pixel stands for a ray at some angle off the optical axis, the lens model says where a ray at that angle lands on the image, and the picture is resampled accordingly.

> **Before you rely on this:** the projection maths is verified numerically. The GLSL is measured against an independent C++ implementation of the same maths across 120 combinations of fit, projection, field of view and direction, agreeing to 0.6 of an 8-bit level (which is the quantisation of the test ramp, not the maths), and the fish/defish round trip lands within 0.001 of a level at moderate settings. It's **since been run inside Resolume on real content**.
>
> Still open: **performance has never been measured**, and although Windows binaries are built and shipped in the release, the plugin's only been used in Resolume on macOS.
>
> This codebase was created with AI assistance, directed and reviewed by a human author.

Installing

Drop the plugin bundle into Resolume's FFGL folder and restart Resolume:

```
macOS    ~/Documents/Resolume Arena/Extra Effects/
         (or /Users/Shared/Resolume Arena/Extra Effects/)
Windows  %USERPROFILE%\Documents\Resolume Arena\Extra Effects\
```

Avenue uses the same layout under its own folder name. Porthole then turns up in the effects browser.

Needs Resolume Arena or Avenue 7.3.1 or newer.

The macOS builds are **Developer ID-signed and notarised**, so the bundle just loads, with nothing to clear and no `xattr` step. The Windows builds aren't code-signed, but plugin files aren't gated the way `.exe` files are, so Resolume loads them normally; only the installer trips SmartScreen, once: **More info** → **Run anyway**.

OpenFX hosts (Resolve, Vegas, Nuke, Natron)

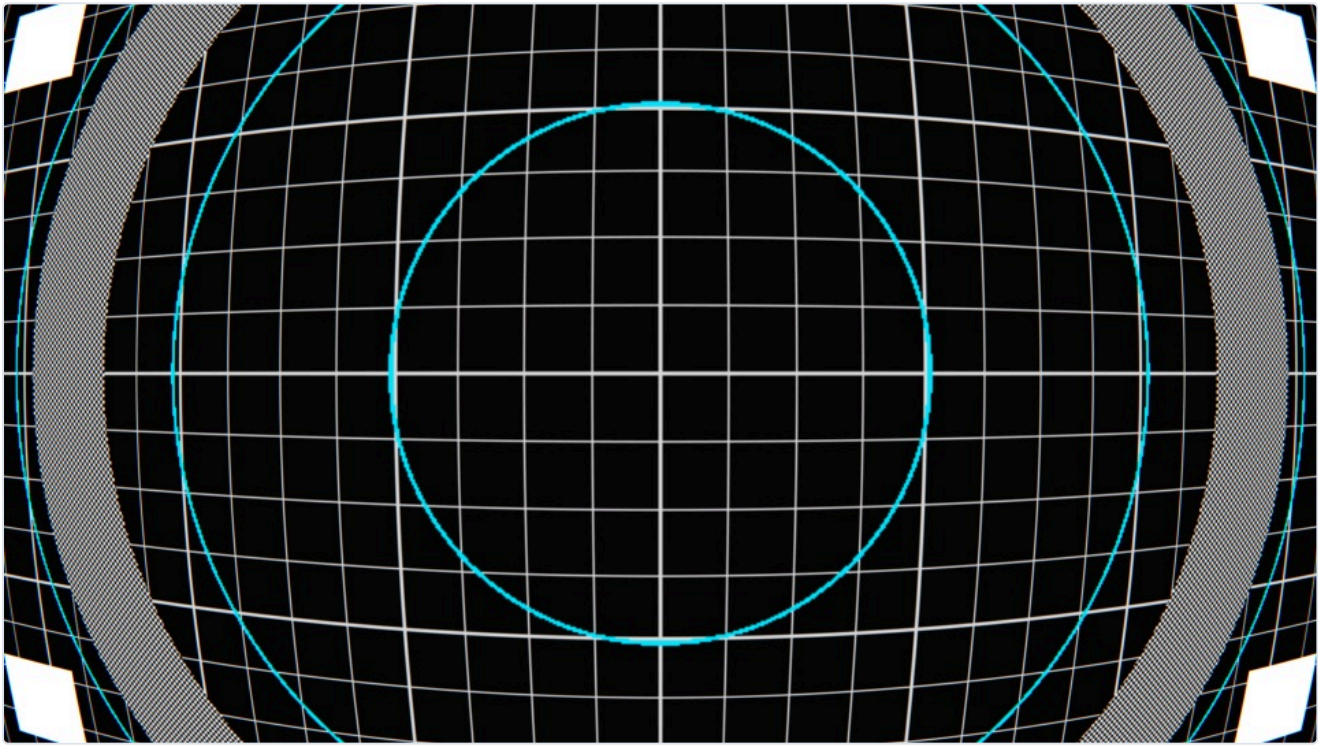
Porthole also comes as an OpenFX plugin, with the same effect and the same controls. Copy `Porthole.ofx.bundle` from the `-ofx-` download into the OpenFX folder and restart the host:

```
macOS    /Library/OFX/Plugins/
Windows  C:\Program Files\Common Files\OFX\Plugins\
Linux    /usr/OFX/Plugins/
```

Projection is *which lens*, not how much

This is the control that matters, and the thing to know about it is that it **isn't a strength knob**. It's a single continuous family that passes exactly through all five projections that have names:

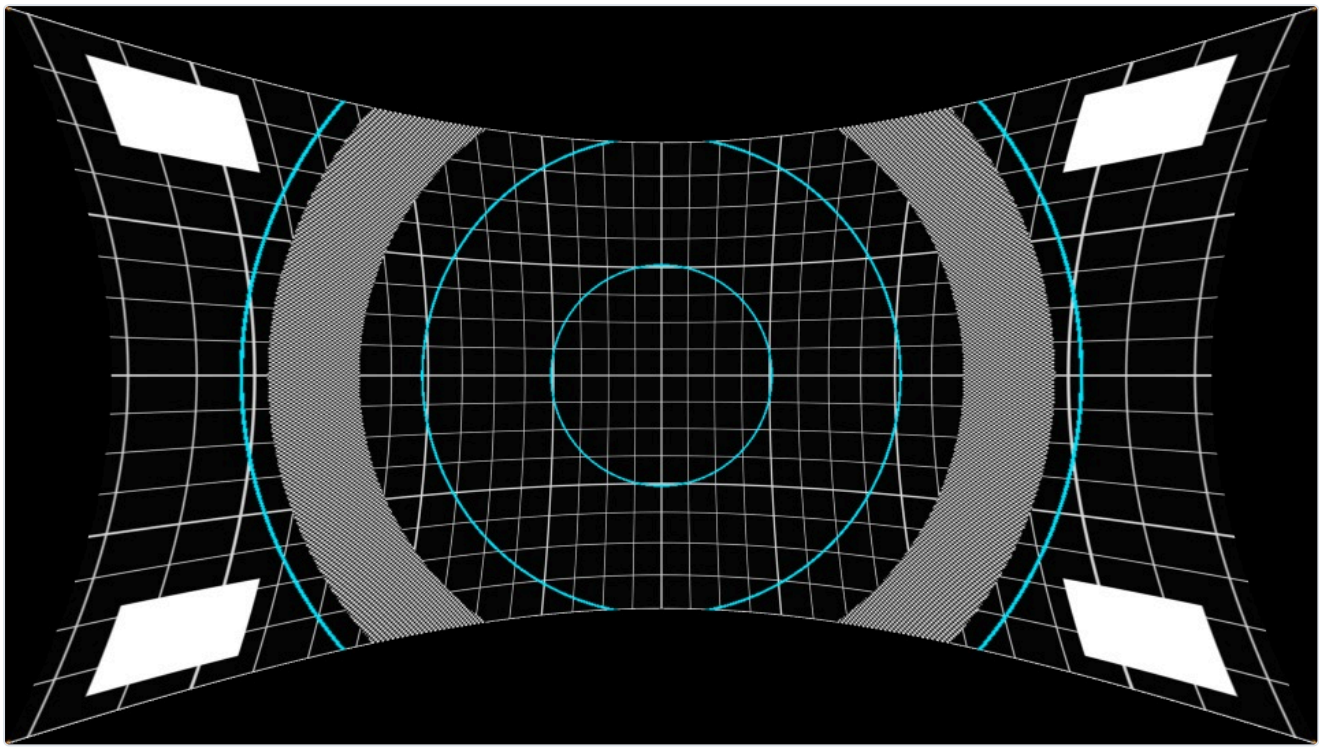
Projection	Formula	What it is
Rectilinear	`r = tan θ`	An ordinary lens — and the identity
Stereographic	`r = 2 tan(θ/2)`	"Little planet"; preserves angles
Equidistant	`r = θ`	The classic fisheye
Orthographic	`r = sin θ`	The mirror-ball



Three behaviours follow from modelling it this way rather than fitting a curve that looks about right. All three are **consequences, not features**, which is why they're exact:

- **Rectilinear does nothing at all**, at any field of view. A flat picture re-photographed through a flat lens is the same picture; the maths says so without being told. If you park the Projection control there and see any change, something is wrong.
- **Defish exactly undoes fish.** It's the same formula with source and destination swapped, not a similar-looking inverse curve.

- **The frame always stays full.** The reference radius is a fixed point of the map, so only the interior is redistributed. Strength changes the *character* of the warp instead of shrinking the picture into a black field.



Undoing a fisheye needs picture from beyond the frame, and there isn't any, so the corners go transparent rather than being made up.

Field of View is the null

There's **no wet/dry mix**, on purpose. Cross-fading two different geometries double-exposes the picture rather than easing between them: you'd see both warps at once, not a blend.

The null is Field of View at zero, where every projection agrees anyway. That's the control to animate if you want the effect to come and go.

Audio reactivity

Resolume can drive any of these sliders from its own audio analysis: click the dropdown beside a parameter, choose **FFT**, and pick a band and gain. No plugin setting is involved.

Porthole's controls happen to be the classic pump targets:

- **Field of View** on the low band is the whole "lens breathes with the kick" move, and because zero is the null, the picture *relaxes to undistorted* between hits rather than sitting in some

second geometry.

- **Chromatic** on hits gives the fringing snap that usually takes a separate effect.
- **Zoom** on overall level, kept subtle, adds the push a camera operator would.

Leave **Projection** alone: it's a slider, but it picks *which lens*, so audio-driving it sweeps between lenses rather than warping one.

Troubleshooting

Symptom	Cause
Plugin doesn't appear in Resolume	Wrong folder, or Resolume older than 7.3.1.
macOS: installed but never loads	The current builds are signed and notarised, but an older download may still be quarantined — and Resolume skips a quarantined plugin silently. `xattr -dr com.apple.quarantine <bundle>`.
Rectilinear is changing the picture	It shouldn't, at any field of view. That is a bug worth reporting.
Corners go transparent when defishing	Correct and deliberate — that picture does not exist. Composite something behind it.
I want to fade the effect in	Animate Field of View to zero rather than looking for a mix control.
Defish doesn't perfectly undo my fisheye footage	It exactly inverts *Porthole's* fish. Real camera glass is not one of these five ideal projections.
Performance is poor	Genuinely unmeasured — please report it with your raster and GPU.

See also

- README: the projection family in pictures, and the download links

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